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Sampling and Low Spatial resolution:

Sampling determines how many pixels are used to represent an image. Low spatial resolution means the image has fewer pixels. Small objects therefore lose important details and appear unclear. Increase the sampling rate by using a higher resolution camera. This produces more pixels and improves object details.

2) Dark and noisy night image:

The image can be improved using image enhancement and noise filtering. Histogram equalisation can increase the brightness and contrast. A median filter can reduce noise while preserving important edges. These techiques make objects more visible in the dark image. Thus, the surveillance image becomes clearer and easier.

3) Quantisation:

It converts continuous intensity values into a fixed number of digital levels. For example, an image can use 256 intensity levels from 0-255. Each continuous intensity value is assigned to the nearest digital levels. This allows the image to be stored & processed digitally. Thus, quantisation is essential for digital image formation.

4) Uneven lighting in industrial inspection:

Histogram equalisation can be used to improve image contrast. It redistributes the intensity values over a wider range. Dark defect regions become more visible against

the product surface. This improves the contrast between defects and normal areas. Therefore, small defects can be detected more easily.

5) Image segmentation for self-driving cars:
Image segmentation divides an image into meaningful regions or objects. Roads, vehicles, pedestrians and other objects can be separated. Each region can be identified using different labels or classes. This helps the vehicle understand its surroundings. Therefore, segmentation supports safe navigation and decision-making.

6) Edge detection for crack detection:
Edge detection identifies sudden changes in intensity in an image. Techniques such as the Canny edge detector can be used. Cracks usually appear as thin edges or intensity changes. Thus, defects can be automatically identified during inspection.

7) Filtering medical image noise:
A median filter is suitable for removing noise from medical images. It replaces each pixel with the median value of its neighbouring pixels. It reduces noise while preserving important edges. This makes the tumor region easier to distinguish. Therefore, filtering improves image quality for medical analysis.

traffic vehicles under different lighting:
object detection using YOLO1 Background subtraction and Thresholding. The importance is it automatically detects counts and track vehicles despite lighting changes. This helps in traffic flow analysis, violation detection and signal control in traffic management. Therefore, it supports efficient traffic management and safety.

9) Motion Blur in camera images:
The blur is mainly caused by object or camera movement during image capture. This movement produces motion blur across the image. Image restoration techiques such as deconvolution can reduce the blur. A suitable motion-blur model can be used during restoration. This improves sharpness and makes objects clear. It estimates the motion blur Kernel (PSF) and reverse it effect to get a sharp image.

10) Image classification for quality inspection:
Image classification assigns a product image to be predefined class. The classes can be defective and non-defective. A trained classification model learns visual features from product images. It can automatically identify defective products during production. Thus, classification enables fast and consistent automated quality inspection.